THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of "Canvas"

Exercises

Self-Study Curriculum

1. The set of all polynomials of degree at most 3 forms a vector space. What is its dimension? Give two different bases. Express the polynomial $2x^3 - x + 5$ in each basis. What changed? What didn't?

2. In your data science work, you routinely choose representations: log-transform, standardization, one-hot encoding. For each of these, describe what vector space the data lives in *before* and *after* the transformation. Which structural properties are preserved? Which are changed?

3. You have a dataset with 100 features. PCA tells you that 95% of the variance is explained by 3 principal components. In terms of this chapter: what just happened to the vector space, the basis, and the dimension? What was the old canvas? What is the new canvas? What was lost? What was gained?

4. A physicist says: "The laws of physics must be the same in every coordinate system." Translate this into vector space language using the concepts from this chapter.